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Predictably Unequal? The Effects of Machine Learning on Credit Markets

Contents

1 Big picture

- **Question.** What happens to the distribution of credit-market outcomes when lenders move from traditional prediction models to more flexible machine-learning models?
- **Setting.** U.S. fixed-rate mortgages originated over 2009–2013, matched between HMDA and McDash, with defaults tracked for up to three years.
- **Core theoretical idea.** A better statistical technology lowers MSE and makes predicted default probabilities more dispersed. This creates winners and losers.
- **Main result.** Black and White Hispanic borrowers are less likely to gain from machine learning relative to White non-Hispanic and Asian borrowers.
- **Mechanism.** Unequal effects can arise from flexibility, triangulation, or both. In this application, flexibility is the main source of unequal effects.
- **Equilibrium result.** Random Forest slightly expands acceptance but increases between-group and within-group rate dispersion.

2 Incremental contribution of the paper

- The paper moves beyond asking whether machine learning predicts better and asks who gains or loses from better prediction.
- It formalizes the mean-preserving-spread logic of improved statistical technology.
- It separates two mechanisms: flexibility in learning nonlinear relationships and triangulation of hidden group identities.
- It uses a large administrative mortgage data set containing race, ethnicity, loan characteristics, and default outcomes.
- It links predicted default probabilities to equilibrium acceptance rates and interest rates.
- It provides a financial-market fairness framework in which input restrictions alone may not prevent unequal outcomes.

Interpretation. The incremental contribution is the connection between algorithmic prediction, fair lending, and equilibrium credit-market allocation.

3 Data construction and measurement

3.1 HMDA-McDash matched mortgage data

HMDA provides borrower race, ethnicity, gender, income, loan amount, loan purpose, and origination information. McDash provides servicing and performance information, including FICO, LTV, interest rates, loan term, documentation status, and delinquency outcomes. The matched sample focuses on 2009–2013 originations and follows delinquency outcomes through 2016.

Interpretation. HMDA supplies protected-group identifiers, while McDash supplies the risk variables and default outcomes needed for prediction.

3.2 Sample restrictions and outcome

The full HMDA originations universe contains 42.2 million loans. The matched HMDA-McDash sample with the seasoning restriction contains 16.84 million loans. After all filters, the main sample contains 9.37 million loans. The outcome is serious delinquency: 90 days or more delinquent at any point during the first three years after origination. Only 0.74% of loans default.

Interpretation. Rare defaults make precision-recall and probability calibration important.

3.3 Borrower groups and covariates

The main groups are Asian, Black, White Hispanic, White Non-Hispanic, Native American/Alaskan/Hawaii/Pacific Islander, and Unknown. The baseline Logit uses linear income, LTV, and FICO. The Nonlinear Logit uses bins for income, LTV, and FICO. Common covariates include origination amount, documentation, occupancy, jumbo status, coapplicant, purpose, term, funding source, mortgage insurance, state, and year.

Interpretation. The comparison holds the permissible information set broadly fixed while changing functional flexibility.

4 Models and empirical specifications

4.1 Ideal predictor under a statistical technology

$$\widehat{P}(x \mid M) = \arg\min_{f \in M} E[(f(x) - y)^2].$$

A better technology has a larger feasible function class, $M_1 \subset M_2$.

Interpretation. Machine learning is modeled as a more flexible class of functions than Logit.

4.2 Mean-preserving spread result

If $M_1 \subset M_2$, then $\widehat{P}(x \mid M_2)$ is a mean-preserving spread of $\widehat{P}(x \mid M_1)$.

Interpretation. Better prediction spreads risk estimates across borrowers, creating winners and losers.

4.3 Flexibility and triangulation

- **Flexibility:** the new technology better learns nonlinear mappings from permissible characteristics to default.
- **Triangulation:** the new technology infers omitted protected-group information from nonlinear combinations of permissible characteristics.

Interpretation. The same unequal effects can be produced by either mechanism, but their regulatory implications differ.

4.4 Default prediction models

$$\Pr(Default_i = 1 \mid x_i) = \Lambda(\alpha + \beta' x_i)$$

for the Logit; a binned version for the Nonlinear Logit; and a calibrated Random Forest for the machine-learning model. Performance is evaluated out of sample using ROC-AUC, precision, Brier Score, and R^2.

4.5 Equilibrium credit-market mapping

The paper solves for a break-even rate $R(x)$ satisfying

$$N(x, R(x)) = 0,$$

where N is the lender's expected net payoff. Borrowers are rejected when no break-even rate exists.

Interpretation. This maps default predictions into acceptance and pricing outcomes.

5 Identification and validation strategy

- The paper identifies predictive and distributional consequences of changing model classes, not a randomized treatment effect of machine-learning adoption.
- Out-of-sample validation uses a train/test split and a separate calibration sample for Random Forest probabilities.
- Race and ethnicity are omitted in the main default prediction models, then added only for diagnostic decompositions.
- Robustness checks vary covariates, samples, and model flexibility.
- Flexibility and triangulation are bounded by comparing performance gains from race controls, race interactions, and Random Forest technology.
- Equilibrium counterfactuals are interpreted cautiously because interest rates are not randomly assigned and denied borrowers are unobserved.

6 Tables and figures

6.1 Figure 1: Unequal effects of better technology + Figure 2: Triangulation

Read: Figure 1 illustrates how flexible nonlinear prediction can penalize the group with more dispersion in a permissible characteristic. Figure 2 illustrates triangulation: nonlinear functions of permissible variables can infer hidden group identity.

Interpretation. These figures show why excluding race from model inputs does not guarantee equal outcomes.

Economic message. Better technology can generate unequal effects through either better learning of observables or indirect inference of group identity.

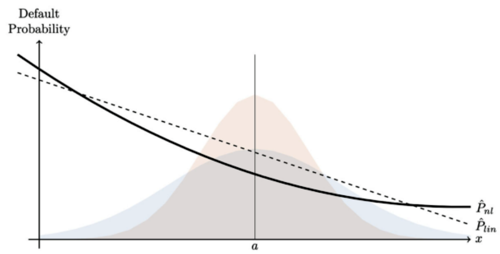

Figure 1. Unequal effects of better technology. (Color figure can be viewed at wileyonlinelibrary.com)

(a) Unequal effects of better technology

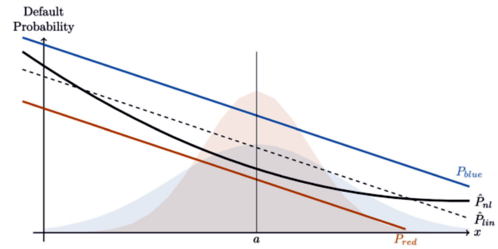

Figure 2. Triangulation. (Color figure can be viewed at wileyonlinelibrary.com)

(b) Triangulation

6.2 Table I: Descriptive Statistics, 2009 to 2013 Originations + Table II: Variables

Read: Table I reports group differences in FICO, income, loan amount, rates, SATO, and default. Table II lists the variables used by the Logit, Nonlinear Logit, and common controls.

Interpretation. The groups differ in the distributions of permissible variables, which is precisely what flexible prediction uses.

Economic message. The data environment is suitable for studying distributional consequences of algorithmic prediction.

Table I
Descriptive Statistics, 2009 to 2013 Originations

This table reports summary statistics for different demographic groups. Income and loan amount are measured in thousands of USD. SATO stands for "spread at origination" and is defined as the difference between a loan's interest rate and the average interest rate of loans originated in the same calendar quarter. Default is defined as being 90 days or more delinquent at some point over the first three years after origination. Data source: HMDA-McDash matched data set of fixed-rate mortgages originated over 2009 to 2013.

Group		FICO	Income	LoanAmt	Rate (%)	SATO (%)	Default (%)
Asian	Mean	764	122	277	4.24	−0.07	0.42
(*N*=574,812)	Median	775	105	251	4.25	−0.05	0.00
	SD	40	74	149	0.71	0.45	6.49
Black	Mean	735	91	173	4.42	0.11	1.88
(*N*=235,673)	Median	744	76	146	4.50	0.12	0.00
	SD	58	61	109	0.71	0.48	13.57
White Hispanic	Mean	746	90	187	4.36	0.07	0.99
(*N*= 381,702)	Median	757	73	159	4.38	0.07	0.00
	SD	52	63	115	0.71	0.47	9.91
White Non-Hispanic	Mean	761	110	208	4.33	−0.00	0.71
(*N*=7,134,038)	Median	774	92	178	4.38	0.02	0.00
	SD	45	73	126	0.69	0.44	8.37
Native Am., Alaskan,	Mean	749	97	204	4.39	0.04	1.12
Hawaii/Pac Isl	Median	761	82	175	4.45	0.04	0.00
(*N*=59,450)	*SD*	51	65	123	0.70	0.46	10.52
Unknown	Mean	760	119	229	4.38	0.00	0.79
(*N*=984,310)	Median	773	100	197	4.50	0.02	0.00
	SD	46	78	141	0.68	0.44	8.85

HMDA). We also verify the robustness of our empirical results to a number of

(a) Descriptive statistics

Table II
Variables

This table lists the variables used in the main models. Section III.E considers additional specifications. Data source: HMDA-McDash matched data set of conventional fixed-rate mortgages.

Logit	Nonlinear Logit
Applicant Income (linear)	Applicant Income (25k bins, from 0 to 500k)
LTV Ratio (linear)	LTV Ratio (five-point bins, from 20% to 100%; separate dummy for LTV=80%)
FICO (linear)	FICO (20-point bins, from 600 to 850; separate dummy for FICO<600)
(with dummy variables for missing values)	
Common Covariates	
Origination Amount (linear and log)	
Documentation Type (dummies for full/low/no/unknown documentation)	
Occupancy Type (dummies for vacation/investment property)	
Jumbo Loan (dummy)	
Coapplicant Present (dummy)	
Loan Purpose (dummies for purchase, refinance, home improvement)	
Loan Term (dummies for 10, 15, 20, and 30 year terms)	
Funding Source (dummies for portfolio, Fannie Mae, Freddie Mac, other)	
Mortgage Insurance (dummy)	
State (dummies)	
Year of Origination (dummies)	

We estimate the model in two ways, varying how the covariates in x enter the right-hand side. In the first model, all of the variables in x (listed in Table II)

(b) Variables

6.3 Figure 3: ROC and precision-recall curves + Table III: Performance predicting default

Read: Figure 3 shows that RF is slightly above Logit and Nonlinear Logit in ROC and precision-recall space. Table III quantifies RF's performance advantage.

Interpretation. Predictive gains are small visually but meaningful in a rare-default market with millions of loans.

Economic message. Accuracy improvements can coexist with unequal distributional impacts.

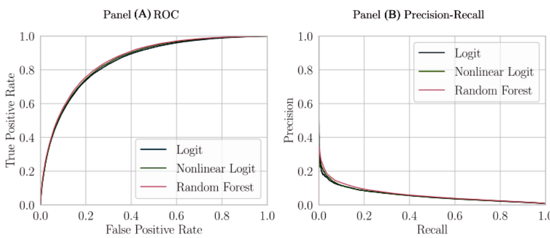

Figure 3. ROC and precision-recall curves. (Color figure can be viewed at wileyonlinelibrary.com)

(a) ROC and precision-recall curves

Table III

Performance of Different Statistical Technologies Predicting Default

This table shows performance metrics of different models. For ROC AUC, Precision Score, and R^2, higher numbers indicate higher predictive accuracy; for the Brier Score, lower numbers indicate higher accuracy. In columns (1), (4), (7), and (10), race indicators are not included in the prediction models; in the other columns, they are included. In columns (2), (5), (8), and (11), the Logit models include the race indicators as simple dummy variables, while in columns (3), (6), (9), and (12), separate Logit models are estimated for each race group (which is equivalent to fully interacting the race indicators with all other variables).

	ROC AUC			Precision Score			Brier Score × 100			R^2		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Model	No Race	Race	Race Int.	No Race	Race	Race Int.	No Race	Race	Race Int.	No Race	Race	Race Int.
Logit	0.8486	0.8491	0.8499	0.0579	0.0582	0.0589	0.7181	0.7179	0.7173	0.0233	0.0234	0.0243
Nonlinear Logit	0.8537	0.8541	0.8543	0.0590	0.0593	0.0600	0.7149	0.7148	0.7147	0.0275	0.0277	0.0279
Random Forest	0.8602	0.8610		0.0620	0.0622		0.7120	0.7118		0.0315	0.0318	

(b) Performance predicting default

6.4 Table IV: Performance predicting race + Figure 4: Example of predicted default probabilities across models

Read: Table IV shows RF predicts race/ethnicity better than the logistic models using permissible variables. Figure 4 shows nonlinear predicted-default contours over income and FICO.

Interpretation. Permissible variables encode some group information, and flexible models exploit nonlinear combinations of those variables.

Economic message. Indirect group inference can occur even without direct use of race.

Table IV

Performance of Different Statistical Technologies Predicting Race

This table shows performance metrics of different models. For ROC AUC, Precision Score, and R^2, higher numbers indicate higher predictive accuracy; for the Brier Score, lower numbers indicate higher accuracy.

Model	ROC AUC	Precision Score	Brier Score × 10	R^2
Logit	0.7436	0.1837	0.5792	0.0607
Nonlinear Logit	0.7442	0.1857	0.5785	0.0619
Random Forest	0.7485	0.1983	0.5738	0.0694

(a) Predicting race

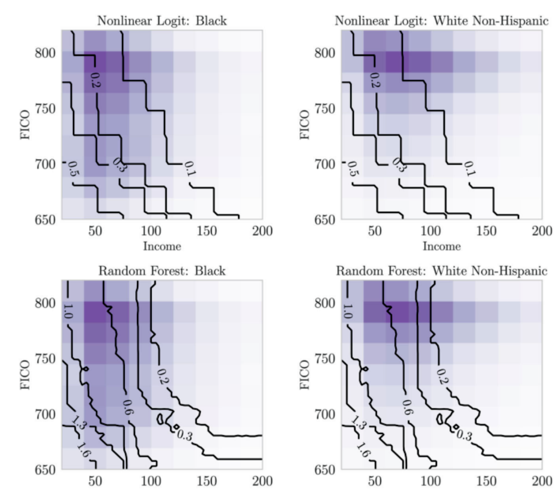

Figure 4. Example of predicted default probabilities across models. This figure shows level sets of default probabilities (in %) predicted from different statistical models for different values of borrower income and FICO (holding other characteristics fixed as explained in text). NL predictions are shown in the top row; RF predictions are in the bottom row. Heatmaps show

(b) Predicted default probabilities across models

6.5 Figure 5: Comparison of predicted default probabilities across models, by race groups + Table V: Alternative models and subsamples of data

Read: Figure 5 compares RF and Nonlinear Logit predictions by group. Table V checks robustness across covariates and samples.

Interpretation. The unequal distributional result is not driven by one covariate or one sample restriction.

Economic message. Black and White Hispanic borrowers are less likely to benefit from the move to RF.

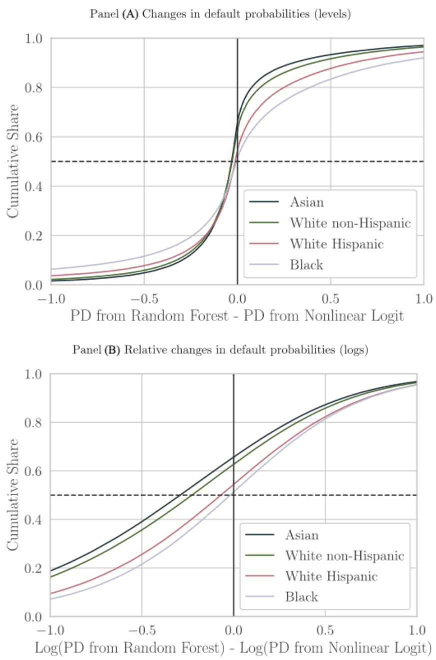

Figure 5. Comparison of predicted default probabilities across models, by race groups.
(Color figure can be viewed at wileyonlinelibrary.com)

(a) Predicted default probabilities by race groups

Table V
Alternative Models and Subsamples of Data

This table shows improvements in performance metrics when going from Nonlinear Logit to Random Forest (Panel A), the percentage of borrowers (by race group) who are assigned a lower PD by the Random Forest than by the Nonlinear Logit (Panel B), and the standard deviation (by race group) of changes in PD when going from Random Forest to Nonlinear Logit (Panel C), for different model versions or estimation samples, as described in the text. The last two rows show for each column the number of observations used for the estimation of the model and the number of observations used for out-of-sample evaluation (testing).

		Change Covariates		Alternative Samples				Change Spec.
	Baseline (1)	Int. Rates (2)	No Fico (3)	No Unknowns (4)	2009 to 2011 (5)	White (6)	GSE+Full Doc (7)	More Interactions (8)
Panel A: Percentage Difference in Performance Statistics between Random Forest and Nonlinear Logit								
ROC	0.77	0.76	1.64	0.73	0.62	0.60	0.58	0.79
Precision	5.06	4.83	11.57	4.33	1.34	3.36	1.56	2.70
Brier Score	−0.40	−0.45	−0.30	−0.41	−0.40	−0.35	−0.30	−0.66
R^2	14.30	15.64	21.28	14.78	12.06	13.71	11.74	25.66
Panel B: Percent Share of Borrowers Who Look More Creditworthy in Random Forest versus Nonlinear Logit								
White Non-Hisp.	12.68	13.07	5.59	12.68	12.69	11.60	9.33	7.10
Asian	15.62	16.14	5.09	15.76	15.44	–	12.00	8.57
Black	1.29	1.04	−0.49	1.48	1.58	–	1.50	−1.44
White Hispanic	4.40	3.77	−0.00	4.54	6.47	–	2.91	−0.30
Panel C: Standard Deviation of Δ Probability of Default between Random Forest versus Nonlinear Logit								
White Non-Hisp.	0.0071	0.0072	0.0057	0.0071	0.0082	0.0073	0.0061	0.0067
Asian	0.0061	0.0061	0.0052	0.0062	0.0074	–	0.0057	0.0057
Black	0.0136	0.0139	0.0084	0.0137	0.0161	–	0.0120	0.0122
White Hispanic	0.0102	0.0103	0.0069	0.0104	0.0128	–	0.0089	0.0092
# Obs. used for estimation	4,591,292	4,591,292	4,591,292	4,108,819	2,757,163	3,495,156	2,522,670	4,591,292
# Obs. used for testing	2,810,996	2,810,996	2,810,996	2,515,826	1,687,444	2,140,187	1,545,697	2,810,996

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(b) Alternative models and subsamples

6.6 Table VI: Decomposition of performance improvement + Table VII: Equilibrium outcomes

Read: Table VI attributes most performance gains to technology/flexibility. Table VII shows acceptance rises slightly but rate dispersion increases under Random Forest.

Interpretation. Better screening can expand access and worsen pricing dispersion at the same time.

Economic message. The main policy issue is not only mortgage approval, but also how risk-based pricing becomes more dispersed within and across groups.

Table VI

Decomposition of Performance Improvement

This table shows the decomposition of improvement for different model performance metrics, as described in the text.

Panel A: Race Controls First			
	Race	Race Int.	Technology
ROC-AUC	6.28	2.04	91.69
Precision	9.05	22.43	68.52
R^2/Brier Score	3.37	4.39	92.24

Panel B: New Technology First		
	Technology	Race
ROC-AUC	89.77	10.23
Precision	94.14	5.86
R^2/Brier Score	92.95	7.05

Consider the performance of the NL and RF models reported in the second and third rows of Table III. If we take the NL without race as a baseline scenario, the greatest increase in predictive power relative to this model will clearly be achieved by both employing a better technology and allowing for the use of race, that is, by using the RF with race. For example, for the R^2 reported in the final two columns of the table, this performance improvement is about 15.6% ($0.0318-0.0275$

(a) Decomposition of performance improvement

Table VII

Equilibrium Outcomes

This table shows predicted acceptance rates, average interest rates (expressed as "SATO" or spread at origination), and the standard deviation in SATO for different groups under the assumption that the statistical model used for default prediction is the Nonlinear Logit (NL) or the Random Forest (RF).

	Accept (%)		Mean SATO (%)		*SD* SATO (%)	
	(1) NL	(2) RF	(3) NL	(4) RF	(5) NL	(6) RF
Asian	92.4	93.3	−0.108	−0.123	0.274	0.322
White Non-Hispanic	90.3	91.1	−0.083	−0.090	0.296	0.356
White Hispanic	85.6	86.4	−0.031	−0.008	0.333	0.414
Black	77.7	79.3	0.022	0.060	0.365	0.461
Other	88.9	89.5	−0.083	−0.088	0.296	0.360
Population	89.8	90.7	−0.081	−0.086	0.298	0.360
Cross-group *SD*	2.165	2.098	0.020	0.029		

We find that the proportion of borrowers in the population that are accepted

(b) Equilibrium outcomes

7 Interpretive synthesis

- Machine learning can improve predictive efficiency while creating unequal gains and losses.
- The unequal effects largely arise from flexibility, meaning that standard input bans may be insufficient.
- The extensive and intensive margins differ: acceptance improves slightly, but rates become more dispersed.
- Fair-lending evaluation should examine both accuracy and distributional consequences.

8 Short summary

- The paper studies machine learning in mortgage default prediction.
- It uses a 9.37-million-loan matched HMDA-McDash sample.
- Random Forest improves out-of-sample prediction.
- Predicted default probabilities change unevenly by race and ethnicity.
- Flexibility is the dominant source of unequal effects in this application.
- Counterfactual equilibrium outcomes show higher acceptance but more dispersed interest rates.

9 Conclusion

9.1 Core conclusion

Machine learning improves mortgage default prediction but distributes gains unevenly across groups.

9.2 Economic intuition

Better prediction spreads risk estimates. When permissible characteristics differ across groups or proxy for group identity, this spread can produce unequal credit terms.

9.3 Incremental contribution

The paper links algorithmic prediction, distributional fairness, and equilibrium credit-market pricing, and distinguishes flexibility from triangulation.

9.4 Takeaway

Evaluating credit algorithms requires studying both predictive accuracy and the distribution of resulting market outcomes.